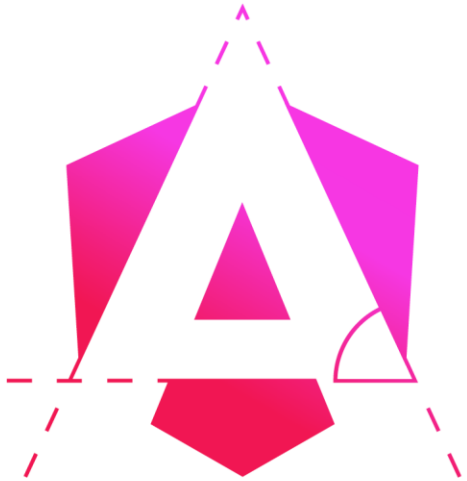


ANGULAR
ARCHITECTS

Angular Performance

Alexander Thalhammer | @LX_T

About me

ANGULAR
ARCHITECTS

About me …

… my office

About me

Hi, it's me → @LX_T https://alex.thalhammer.name/

- **Alexander Thalhammer** from Graz, Austria (est. 1983)
  - **Angular Software Tree** GmbH (est. 2019)
- Web Dev for 24 years (I‘ve come a long way baby)
- 2011 - 2017 WordPress Dev (Web, PHP & jQuery)
- Since 2017 **Angular Dev** (Web, TS, Rx - NG 4.0.0)
- Since 2020 **Angular Evangelist, Coach** & **Consultant**
  - Member of **Angular Architects** https://www.angulararchitects.io/

ANGULAR ARCHITECTS

About you

ANGULAR
ARCHITECTS

Tell us sth about yourself

- Hi, my name is …
- I currently work on …
- A good performance is …
- I have the following problem …
- My goal(s) for this workshop is (are) …

About you

Let us know about your

- Which OS
- Which browser
- Which IDE (IntelliJ/WebStorm vs VS Code)
  - Make sure to update to latest version!!!
- How big is your display?

Pollings

About your team / projects

– Before I tell you sth about our workshop ☺

Your work

About this workshop

ANGULAR
ARCHITECTS

Every participant should

– understand the ***importance*** of performance

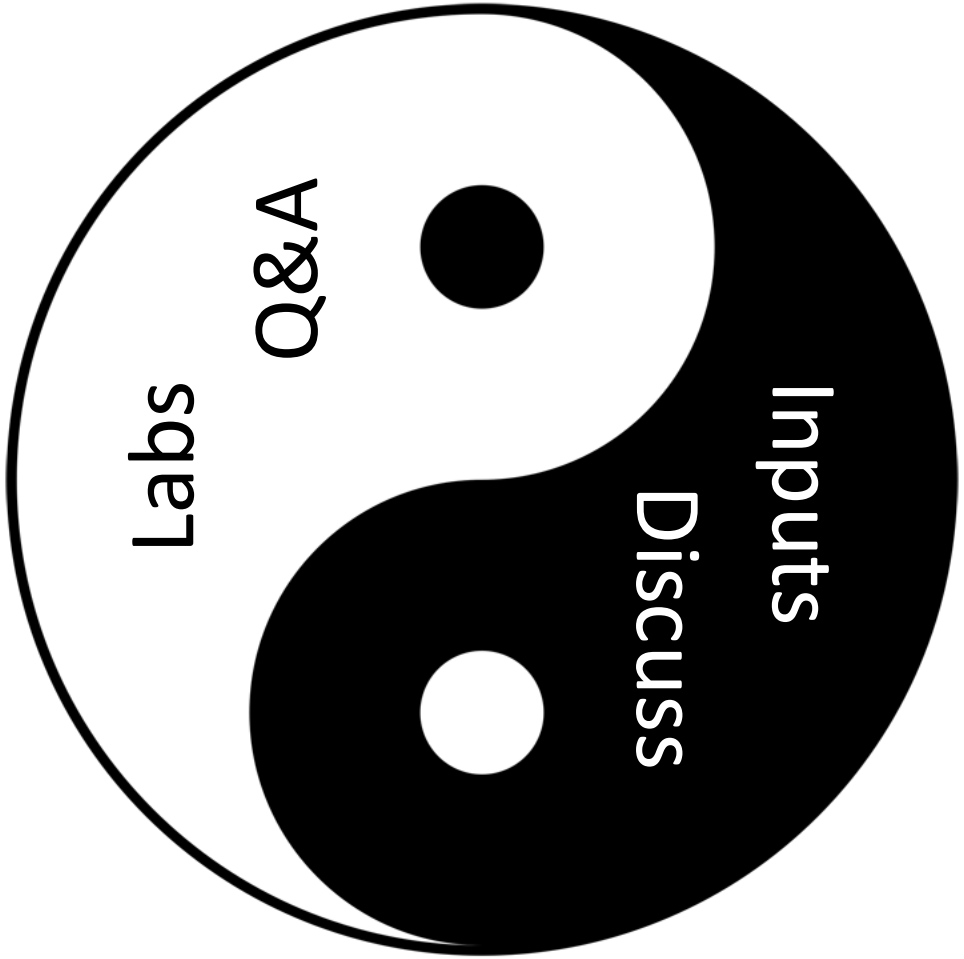
– know how to ***measure*** the performance

– know how to ***tweak*** the performance

– become a ***performance evangelist*** in her/his project/team/company

– have ***fun*** along the way

My goals

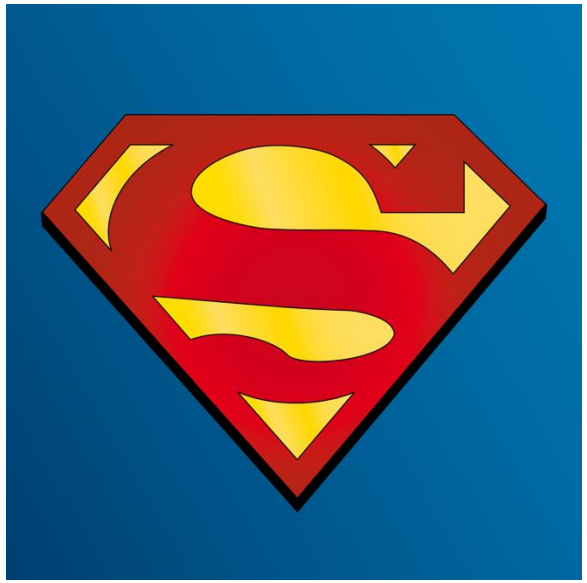


Didactics

We work together I

- Don’t hesitate to interupt me at ***ANY TIME***
  - **Questions**
  - Remarks or
  - Feedback ☺

- Somebody else might have the same **question**
  - Be a hero and ask it



Workshop

Timetable (approx.)

- 09:00 – 12:00 (incl. 15m ☕ break)
- 1hr 🍲 break
- 13:00 – 16:30 (incl. 15m ☕ break)
- Q&A / feedback
- Incl. delay we will end at 17:00 latest 🏁

Workshop

ANGULAR
ARCHITECTS

What is good performance?

– What did you say?

– **Web performance** refers to the speed in which web pages are downloaded and displayed on the user's web browser. **Web performance optimization (WPO)**, or **website optimization** is the field of knowledge about increasing web performance.
-- https://en.wikipedia.org/wiki/Web_performance

Why Performance Optimization?

– According to *Amazon*, 0.1s less loading time results in around 1% more sales

– According to *Google*, 40% of visitors will abandon a website if it takes longer than 3 seconds to load

– Also according to *Facebook*, when people have to wait too long for a webpage to load, they're more likely to abandon the page all together

Angular Performance Optimization

We distinguish between

- Initial load performance (classical web performance)
- Runtime performance (during usage, e.g. scroll frame rate)

Agenda & Labs

ANGULAR
ARCHITECTS

Agenda

- 11: UAAO Framework
- 12: Audit Tools
- 13: Initial Load - Assets & Build
- 14: Initial Load - Lazy Loading & Deferrable Views
- 15: Initial Load - SSR & SSG
- 16: Runtime - Synchronization
- 17: Runtime – Best Practices
- 18: Modern NG (14 – 19)
- 19: Bonus & Conclusions

Labs

00: Getting Started	13: Initial Load - Assets & Build	16: Runtime - Change Detection
11: Web Performance Audit	14: Initial Load - Lazy Loading & Deferring	17: Runtime – Best Practices
12: Build Analysis	15: Initial Load - SSR & SSG	18: Modern NG (14 – 19)

Setup for Labs

- NodeJS version ^18.19.1 or ^**20.11.1** or ^22
  - https://angular.dev/reference/versions#actively-supported-versions

Actively supported versions

This table covers Angular versions under active support.

Angular	Node.js	TypeScript	RxJS
19.0.x	^18.19.1 || ^20.11.1 || ^22.0.0	>=5.5.0 <5.7.0	^6.5.3 || ^7.4.0
18.1.x || 18.2.x	^18.19.1 || ^20.11.1 || ^22.0.0	>=5.4.0 <5.6.0	^6.5.3 || ^7.4.0
18.0.x	^18.19.1 || ^20.11.1 || ^22.0.0	>=5.4.0 <5.5.0	^6.5.3 || ^7.4.0
17.3.x	^18.13.0 || ^20.9.0	>=5.2.0 <5.5.0	^6.5.3 || ^7.4.0
17.1.x || 17.2.x	^18.13.0 || ^20.9.0	>=5.2.0 <5.4.0	^6.5.3 || ^7.4.0
17.0.x	^18.13.0 || ^20.9.0	>=5.2.0 <5.3.0	^6.5.3 || ^7.4.0

Try "node -v"

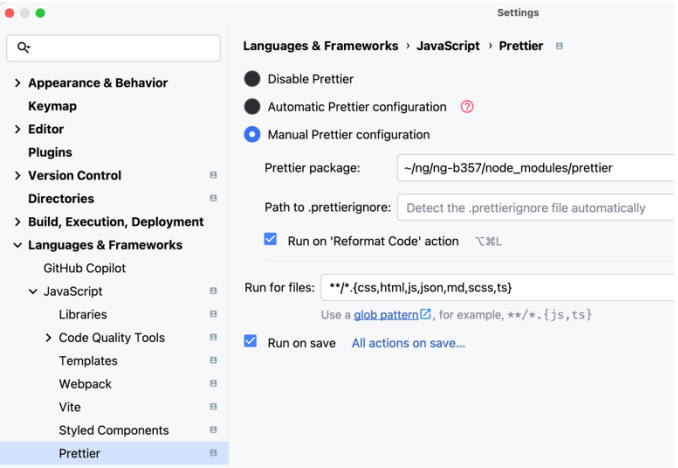
Use NVM ☺

Mac Alternative: Use Homebrew

Setup for Labs

- Git
  - Personally, I used to use Sourcetree
    - but currently migrating to Tower | Fork
- VS Code (free) | IntelliJ/WebStorm (better)
  - Personally, I prefer IntelliJ/WebStorm
- Chrome (or Chromium based alternative)
  - Other browsers are okay for private matters

Recommendations



- Use prettier
  - especially for .ts files
- Use ESLint
  - in your IDE of choice and/or from terminal
- Use Git
  - we recommend to at least commit after every lab
  - easier switching back and forth with my solution

Starter Kit

- Incl. Slides
- Incl. Labs (exercises)
- Clone https://github.com/L-X-T/ng-b357

Ready for takeoff

ANGULAR
ARCHITECTS

Workshop Updates

- updates are always done "on-the-fly"
- NG 19
  - Hybrid Rendering
  - Incremental hydration

Disclaimer

- 00 Intro
- 01 UAAO Framework
- ☕ break
- 02 Audit Tools
  - Labs

Day 1

ANGULAR
ARCHITECTS

- 03 Initial Load - Assets & Build
  - Lab
- ☕ break
- 04 Initial Load – Lazy Loading
  - Lab
- 05 Initial Load – SSR & SSG
  - Lab

Day 2

- 06 Runtime - Synchronization
  - Lab
- ☕ break
- 07 Runtime - Best Practices
  - Lab
- 08 Modern NG (14-19)
  - Lab
- 09 Bonus & Conclusions

Day 3